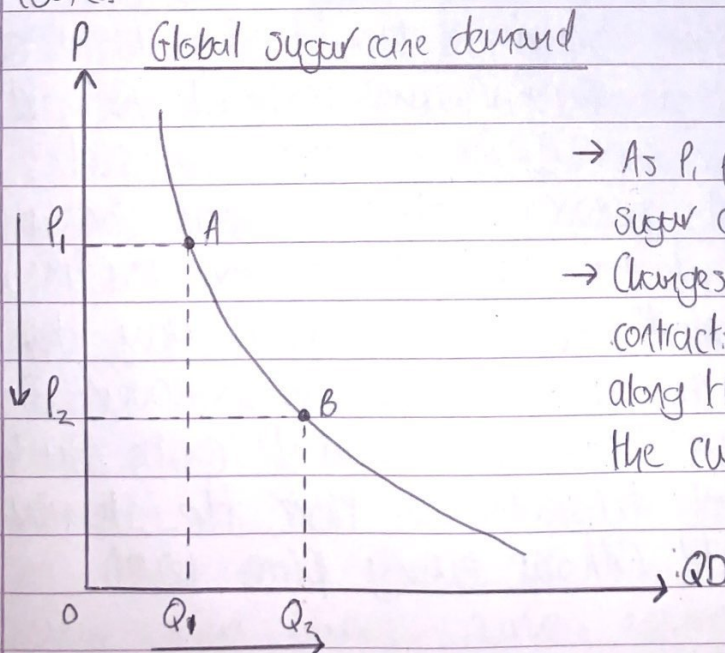


Supervision 1

1. I will use the example of sugar cane crops to provide examples of how economists can categorise the factors affecting supply and demand, thus quantity and price at equilibrium, of a good.

The main determinant of demand for any product is its price. The relationship between a product's price and demand is known as its own-price elasticity, explored further in part 2, and can be observed visually as the slope of the demand curve. The demand curve is almost always downward sloping, meaning that ^{at} higher prices there is less demand, because of diminishing marginal utility. This means that each additional unit of consumption brings less satisfaction, therefore lower prices are required to induce additional consumption. Further, at low prices, more and more consumers in developing nations will be able to afford, thus demand, products derived from sugar canes.

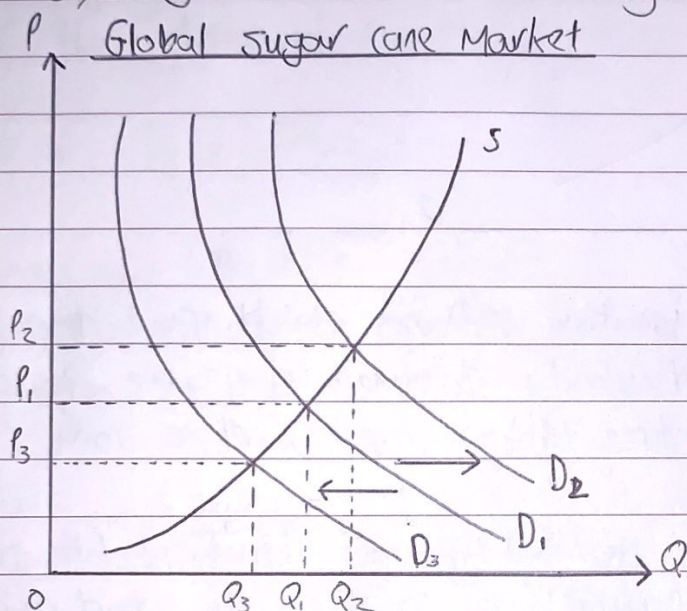


- As P_1 falls to P_2 , more consumers demand sugar canes, so quantity rises to Q_2
- Changes in price lead to expansions or contractions of demand, meaning movement along the curve instead of a shift of the curve

Whilst product price is the most important, there are several other factors which can affect demand, including the price and demand for complementary goods. These are goods which are consumed jointly with sugar cane. For example, sugar cane plants can be used as livestock fodder. Assuming local farmers had no available substitutes for sugar cane, a declining price / increasing quantity of livestock would increase demand for sugar cane at every price level. Additionally, for some products we must consider the effects of derived demand. Sugar cane is used as an input when producing other products such as sugar and rum. As such, if consumers are demanding more of these products, the demand for sugar cane will rise at any given price level.

Conversely, the price and availability of substitute goods to consumers will impact the demand for sugar cane. There is a ^{positive} ~~negative~~ relationship between the price of a substitute and the demand for sugar cane. For example, if artificial sweeteners became cheaper, some consumers would stop using sugar cane inputs to create sugar and instead use artificial sweeteners in their soft drinks, confectionaries etc. Using this same example, we can observe how consumer tastes can affect a product's demand. As more consumers become environmentally conscious, they may develop a greater preference for artificial sweeteners over sugar cane, if there are lower environmental costs such as wasted water and deforestation. Here, the demand for sugar cane would fall at every price level.

Finally, the income trends of consumers can affect the demand for a product. The relationship between income and demand can be positive or negative, as explored later, but given the nature of sugar as a derivative of sugar cane, sugar cane is likely a normal necessity good.

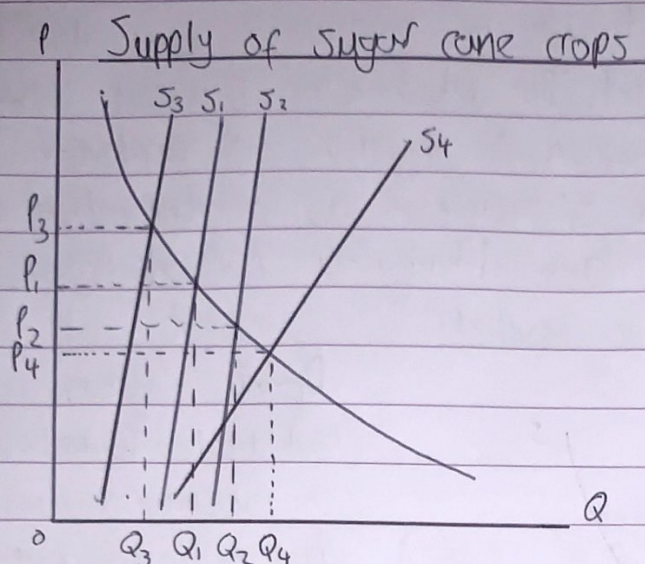


$D_1 \rightarrow D_2$: increase in price of substitutes (sweeteners), increase in consumer incomes, decrease in price of complements or derived goods

$D_1 \rightarrow D_3$: inverse of above

In order to understand market equilibrium, economists must also understand the determinants of the supply of a good. Supply is upward-sloping because, over time, producers respond to the incentive of higher selling price by expanding capacity - this represents a move along the supply curve. Further, as output increases firms face higher marginal costs, thus require higher prices to justify production.

It is important to note the peculiarities of supply with respect to sugar cane, and most other crops. In the short run, that is within one season, supply is extremely inelastic as once seeds are planted, the maximum supply of crops from those fields are fixed. This results in a steep supply curve, meaning shifts in demand and supply have a magnified effect on price.



In the short run, weather patterns could affect crop yields. For example, a particularly extreme hurricane season in the Caribbean could reduce global crop supplies from S_1 to S_3 .

As mentioned early, seasonality and land available can make supply inelastic and unable to shift in the short run. However, the develop of new technologies could both increase supply and make it more elastic. For example, the widespread adoption of high-tech greenhouses could expand supply to S_4 by increasing the availability to arable land and allowing farmers to respond to year round price fluctuations.

Finally, the supply of a product can be affected by interventionist measures such as taxes and subsidies aiming to incorporate externalities into market equilibrium. Many agricultural product goods such as sugar cane are subject to heavy and consistent subsidies by national governments, in order to increase food security. Such programmes, including the EU's CAP, increase supply.

2.

- (a) Own-price elasticity is a measure of the responsiveness of a product's demand to a change in its price. It is calculated by multiplying the ratio of price to quantity by the slope of the demand function.

$$\epsilon = P/q \times \Delta q / \Delta p$$

Cross-price elasticity follows the same intuition, but measures the responsiveness of demand for good A to a price change in good B. It is calculated by the formula:

$$XED: \frac{\Delta q_x / q_x}{\Delta p_y / p_y}$$

A positive XED value indicates the goods are substitutes and a negative value indicates they are complements. The absolute value indicates the strength of the relationship, with 0 being none and 1 being perfect.

Income elasticity refers to the responsiveness of a product's demand to changes in consumers' incomes. It's calculated as:

$$YED = \frac{\Delta q / q}{\Delta y / y}$$

- (b) Elasticity is a measure responsiveness, and thus 'inelastic' demand curves are those for products which have a low responsiveness of demand to changes in their own price. Applying the above formula, demand is classed as inelastic

if $|E| < 1$. Products may be price inelastic because they have a high degree of necessity, such as petrol. When petrol prices rise, the scope for demand to fall is limited because people who commute to work in cars will still require petrol. This has implications for the pricing strategies of producers of such goods; it may well be that by increasing prices, firms can increase their revenues, as the money generated from higher prices exceeds the money generated from lower sales. Indeed, this has occurred in the oil industry with the OPEC cartel restricting output to artificially raise prices. However, it is worth noting that elasticity varies over time (and also at different prices). In the long run, products tend to become more elastic. Even though oil is price inelastic, if prices consistently rose, consumers might consider substituting their mode of transport to electric cars, thus more substantially reducing demand.

On the other hand, many goods are price elastic, meaning following a price change, there is a proportionally larger change in demand. These are typically products which have a lower degree of necessity because of the availability of substitutes. For example, if Cadburies increased the price of their chocolate bars, a large proportion of customers would be willing and able to switch to Mars. That said, building a powerful brand could allow firms to decrease the price elasticity of their products. In the confectionery industry, this can be observed through Ferrero Rocher's upmarket branding allowing them to command higher prices.

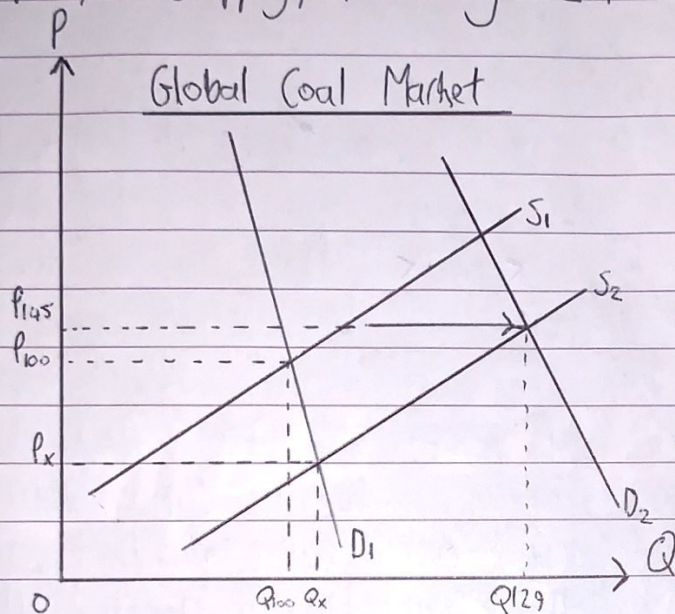
7. c. Economists can categorise goods as normal, luxury or inferior based on their income elasticity of demand.

Normal goods are those which have a positive YED, meaning that as incomes increase so does demand for the product, and vice versa. Within this, economists distinguish between normal necessity goods and luxury goods. Normal necessity goods are those for which $0 < YED < 1$, thus as incomes increase so does demand, but proportionally less. For example, if a consumer's income rose by 20%, they might consume slightly more bottled water, say ~~20%~~ 5% more. They would not consume 50% more, for example, as they would rather spend the additional income on luxury products which bring greater utility. These are luxury goods, those which have a YED of over 1. This means as incomes increase, demand increases by a proportionally greater amount. A typical example would be meals out at high-end restaurants.

Thirdly, some products are labelled "inferior". They have a negative YED, meaning that as incomes rise consumers will demand them less, as they substitute them for more expensive alternatives. Examples include bus travel: as consumers' incomes rise, the increased cost of running a car may become worth the additional comfort and convenience, thus consumers demand less bus travel.

3.

- (a) According to the data, from 1973 to 1980 the price of coal rose by 45%, adjusted for inflation. In the same period, coal output, or supply, rose by 29%.



We can use a supply and demand graph to understand the price and output trends. If S_1 increased to S_2 , *ceteris paribus* we would see an increased quantity, Q_x , and a lower price, P_x . Thus, the price increase observed in the data must be the result of an increase in demand to D_2 .

The likely causes for such an increase demand include an increase the price of substitutes within the energy sector, or an overall increase in global demand for energy as emerging nations industrialise. Here, the former is more likely given the oil crises of the 1970s dramatically increased oil prices. This would push firms to substitute oil inputs for substitutes such as coal where possible, driving a demand increase.

- (b) To confirm: data on oil prices, other energy sources information, breakdown of demand composition.